Patrick Phillips

TS/SCI Clearance

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ACADEMICS

University of Chicago — ***2022 - 2023***

M.S. in Computer Science with Specialization in Data Analytics

- **GPA 3.86/4.0**. Completed while working full-time.
- Relevant Coursework: Causality with Machine Learning, Cloud Computing (AWS), Advanced Algorithms, Unsupervised Learning and Data Analysis, C/C++ for Advanced Programmers, Computer Systems

University of Rochester — ***2017 - 2021***

B.S. in Computer Science, B.S. in Applied Mathematics, B.A. in Physics, B.A. in Engineering Science

- **GPA 3.96/4.00**, Dean's List all eligible semesters, *magna cum laude*.
- New York State Academic Excellence Scholarship, American Council of Engineering Scholarship.
- Relevant Coursework: Machine Learning, Linear Algebra, Artificial Intelligence, Complex Analysis, Discrete Mathematics, Computer Vision, Computational Complexity, Multidimensional Calculus

SKILLS

- Experienced in Python, Ruby on Rails, C# (.NET), Java, C, C++, JavaScript, SQL, AWS, Kubernetes, Selenium, AI/ML, TenorFlow, SciPy, scikit-learn, Pandas, PyTorch, R, MATLAB, Linux/UNIX terminal, Windows Server, Microservices, MongoDB, Jenkins, Docker, CI/CD, Agile, Git.

TECHNICAL EXPERIENCE

General Dynamics Information Technology — (*Remote*)

Principal AI Engineer — ***November 2025 – Present***

- Rapidly prototyping AI native systems and web applications built with a combination of custom models and state of the art VLMs. Areas of focus include deploying AI systems on edge compute, sensor fusion, and integrating AI into secure enterprise workflows.

Sr. Software Developer — ***October 2023 – November 2025***

- Supporting USSOCOM operations as a full stack web developer in an Agile team. Lead migration of Ruby on Rails application suite to Kubernetes using AWS services.

Riverside Research — Centreville, VA (*Remote*)

Fullstack Software Developer — ***January 2021 – October 2023***

- Supporting the Collection Planning Suite, a large-scale web application used for satellite scheduling. Tools I use include C#, .NET, Oracle SQL, JavaScript, HTML, Jira, Agile, Windows Server, and AWS.

AI/ML Scientist I — ***April 2020 – January 2021***

- Investigated applications of AI/ML to operations in space asset planning. Designed and developed a novel optimization scheme for satellite scheduling, with a finished prototype in Python.

Argonne National Laboratory — Chicago, IL

Machine Learning and Quantum Chemistry Internship — ***May 2022 – October 2022***

- Researched Auger spectra decay prediction based on molecular structure in support of the Quantum Chemistry and Molecular Physics group. This work involved implementing and managing large scale deep learning systems in Python.

University of Chicago Rochester, NY
Computer Science Teaching Assistant ***September 2022 – December 2022***

- Lead focused problem sessions and reviewed course materials for graduate level Algorithms (MPCS 55001).

University of Rochester Rochester, NY
Computer Vision and Reinforcement Learning REU ***May 2020 – August 2020***

- Combined concepts from state-of-the-art reinforcement learning, game theory, and meta learning to develop an agent specialized in two player zero sum games.

Computational Complexity Research Internship ***January 2021 – May 2021***

- Researched foundational problems in complexity theory, with a focus on restricted counting classes.

Computer Science Teaching Assistant ***September 2018 – May 2021***

- Lead problem sessions and graded assignments for Design & Analysis of Efficient Algorithms (CS 282), Computational Complexity (CS 286), and Intro to Computer Science (CS 171).

Technical University of Hamburg (TUHH) Hamburg, Germany
DAAD Research Internship ***May 2019 – August 2019***

- Researched the 'informative path planning problem' where an agent aims to efficiently move around a scalar field while taking measurements to quickly learn a Bayesian estimation of the true field. This work used Python simulation as well as an underwater autonomous vehicle.

PERSONAL PROJECT HIGHLIGHTS

- Using Angular, Python, and Firebase hosting, I developed my wife Michaela Chan's art website https://michaelachan.com/.

- Implemented several ML algorithms from scratch including backpropagation, Expectation Maximization for a HMM and for Mixture of Gasusians, K-Nearest-Neighbors, variations of genetic algorithms, and more. These were done in Python and can be found on my Github along with many other projects.
- Completed the projects in *Computer Systems: A Programmer's Perspective: Robert O'Hallaron* as part of my computer organization course.
- Collaborated on many data science projects with specific applications in recommender systems, language processing, medical diagnosis classification and more as a member of the Undergradute Data Science Council.
- Created chess, poker (Texas-Hold 'em), and connect-4 AI players using tree search algorithms with alpha-beta pruning, Monte Carlo search, and other added heuristics specific to the respective games, in Java.
- Using Kotlin, Jetpack Compose, Python, machine learning, diffusion models, and RESTful APIs I developed the Android app Holophone for AI Music generation and AI Music to Video generation. Available on the Google Play Store.

RESEARCH PAPERS

- Patrick A. Phillips, Jing Bi, Jing Shi, Chenliang Xu. "Reinforcement Learning in Two Player Simultaneous Action Games" University of Rochester. (2020) https://arxiv.org/abs/2110.04835

- L. Hemaspaandra, M. Juvekar, A. Nadjimzadah, and Patrick A. Phillips. "Gaps, ambiguity, and establishing complexity-class containments via iterative constant-setting." (2021) https://arxiv.org/abs/2109.14764